# Leveraging Smartphone Inertial Data for User Authentication

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***Abstract*—Walking is a fundamental human activity, and gait patterns exhibit distinctive characteristics across individuals, making them suitable for biometric authentication. Recent advances in mobile technology have enabled the integration of Inertial Measurement Unit sensors in smartphones, allowing unobtrusive user authentication using personal devices. In this work, we propose a gait-based authentication framework leveraging inertial data, including acceleration, angular velocity, and gravity, acquired from a smartphone. The multimodal time series are fused and encoded into a latent representation using an LSTM-based autoencoder. A One-Class Support Vector Machine is then trained on this latent space to model the unique walking patterns of each user. Experimental results demonstrate the effectiveness of the proposed approach and highlight its robustness to variations in walking conditions, including scenarios involving gait alterations due to injuries. These findings support the potential of smartphone-based gait authentication as a practical and reliable biometric solution.**

***Index Terms*—User Authentication, Multi-Modal Time series, Deep learning, Feature Fusion, Gait Analysis, Inertial Data**

## I. Introduction

Human gait is a mode of locomotion characterised by a series of rhythmic movements of the lower extremities that propel the body forward with minimal energy expenditure.

Analysing human gait is essential in various fields, including sports biomechanics [1], personal authentication [2], robotics [3], clinical evaluations, and rehabilitation [4]. Moreover, in recent years, gait-based human authentication has gained significant attention as a biometric method due to the difficulty of replicating an individual's walking style and its potential for unobtrusive use in real-world conditions. Its adoption has been assessed in applications like surveillance [5], security [6] and forensic identification [7].

Common sensors used for gait analysis and authentication include wearable Inertial Measurement Unit (IMU) devices [8], video cameras [9], motion capture systems [10], and ground contact force sensors [11]. IMU sensors, in particular, have been widely adopted for gait-based analysis and authentication, showing compactness and portability. With the widespread use of smartphones, inertial data can now be easily collected using the built-in accelerometer and gyroscope, widening the range of possible tasks. The existing literature enhances the potential of inertial data for effective gait analysis and authentication [12], [13] while showing performance limitations due to data variability and the influence of factors, fatigue, injuries, or carrying conditions [14].

Starting from these considerations, in this paper we investigate the adoption of smartphone inertial data for user authentication. We propose a methodology based on a representation stage, leveraging a Long Short-Term Memory (LSTM)-based mid-fusion Auto-Encoder, and an authentication stage implemented with a One-Class SVM (OCSVM) [15]. When approaching authentication as an anomaly detection problem the One-Class model, compared to multi-class methods, has the potential to easily scale up to new individuals. Our unsupervised representation is achieved analysing a dataset of multiple identities, thereby incorporating a variety of gait traits. Later, such representations are used to feed a personal OCSVM for each identity to be recognized. We provide an experimental analysis to show the robustness of the proposed pipeline on a dataset acquired in-house, including sequences of 50 users walking in different conditions.

Overall, our work represents a feasibility study on gait-based biometric authentication on smartphones, to alert the user when a deviation in the walking signature is detected, hence potentially indicating that the device has been stolen.

## II. Related works

The literature of gait-based human authentication is wide, and the use of different sensors have been explored over the years, including radar-based [16], [17], pressure force sensors [18], [19], and inertial data [20], [21].

With recent technological advancements, IMUs have been ubiquitously integrated into everyday devices such as smartphones, smartwatches, and tablets. The widespread adoption of these devices has facilitated the development of novel methods for unobtrusive gait analysis and authentication, of interest for our work. Since the foundational study in this field appeared in 2005 [22], the literature has extensively explored smartphone-based gait authentication, initially relying heavily on classical machine learning architectures. For instance, Kala et al. [23] evaluated Support Vector Machine and Random Forest classifiers using smartphone acceleration data collected from a user's right pocket. Expanding on this approach, Ehatisham-ul-Haq et al. [24] incorporated accelerometer, gyroscope, and magnetometer data from smartphones placed at multiple body locations, evaluating a broader suite of traditional algorithms including Decision Trees, K-Nearest Neighbors, and Bayesian Networks.

More recently, to capture the complex spatial-temporal dynamics of human gait, the research focus has shifted toward deep learning methodologies. Gadaleta and Rossi [25] proposed a Convolutional Neural Network (CNN) to authenticate users based on accelerometer and gyroscope signals from a smartphone in the front trouser pocket. Similarly, Zou et al. [26] developed a hybrid CNN-LSTM architecture to process inertial data collected 'in the wild.' Addressing the inherent computational constraints of deploying such models on mobile devices, Basabi et al. [27] recently developed a highly efficient CNN-based authentication algorithm that leverages knowledge distillation to compress the model size while maintaining high performance on continuous time-series sensor data.

In contrast to the majority of existing literature, which relies heavily on supervised learning paradigms for feature extraction, our proposed methodology adopts an unsupervised approach. By utilizing an Auto-Encoder to extract latent features, our system effectively learns the unique signature of a user's gait without requiring labelled data. To the best of our knowledge, there are published works on gait authentication using acceleration sensors and gyroscope sensors, but none of them incorporates gravimetry.

## III. Proposed method

We refer to an authentication setting, where an identity $i$ is characterised by a set of $N_i$ gait sequences $S_i = \{\mathbf{x}_{i,k}\}_{k=1}^{N_i}$. Specifically, we formulate the task as an anomaly detection problem, where we are assuming we have a set of training data to be used to compute the model of the identity $i$, and then we use this model at run time to detect any deviation or anomaly (a failed authentication). We design a classical pipeline that learns a representation with an auto-encoder and then uses it as an input for an anomaly detection algorithm based on a One-Class SVM. In this section, we provide details on our pipeline, depicted in Fig. 1.

### A. LSTM-based Auto-Encoder

We first learn a gait encoding with an unsupervised approach. To this purpose, we consider an entire dataset incorporating $N$ identities, $\mathcal{S} = \cup_{i=1}^{N} S_i$, and use it to train an Auto-Encoder solving a reconstruction task.

The Auto-Encoder (see Fig. 1) employs LSTM layers. We assume we have as input a sequence $\mathbf{x} = [x_i, \cdots, x_T]$ of length T, $\mathbf{x} \in \mathbb{R}^{9\times T}$ where each observation $x_t = [a_t, w_t, g_t]$ includes the $x$, $y$, and $z$ instantaneous components of inertial data (acceleration, angular velocity, gravity), hence $a_t, w_t, g_t \in \mathbb{R}^3$. We apply a mid-fusion among the features, which are first processed independently using 3 LSTM layers with hidden states $\mathbf{h}_a, \mathbf{h}_w, \mathbf{h}_g \in \mathbb{R}^{16}$. The concatenated feature vector $\mathbf{h} = [\mathbf{h}_a, \mathbf{h}_w, \mathbf{h}_g] \in \mathbb{R}^{48}$ is used to fed a global LSTM layer, again with a 16-dimensional embedding, which also serves as a final sequence encoding. The decoder structure is designed symmetrically to the encoder.

### B. OCSVM Anomaly Detection

The One-Class Support Vector Machine [28] is a variant of the traditional Support Vector Machine, primarily designed for estimating the support of a high-dimensional distribution. In our scenario, the main goal is to classify through gait an individual as a genuine user or an impostor, where the genuine user is represented by the target distribution, while impostors represent all other unknown individuals (hence anomalies with respect to the target distribution).

We adopt the mathematical formulation in [15] and, given $\ell$ training samples, we minimize the dual problem

$$\min_{\boldsymbol{\alpha}} \frac{1}{2}\boldsymbol{\alpha}^T Q \boldsymbol{\alpha} \tag{1}$$

subject to the constraint $0 \leq \boldsymbol{\alpha}_k \leq 1/(\nu\ell)$ for all $k = 1, \cdots, \ell$ with $\sum_k \boldsymbol{\alpha}_k = 1$ and $Q_{kj} = K(\boldsymbol{v}_k, \boldsymbol{v}_j)$, being $K$ a kernel function. The decision function is

$$\hat{f}(\boldsymbol{v}) = sign\left(\sum_{k=1}^{\ell} K(\boldsymbol{v}_k, \boldsymbol{v}) - \rho\right) \tag{2}$$

A OCSVM model is trained for each identity to be authenticated. The hyperparameters of the model ($\nu$, $\rho$ and the kernel parameter $\gamma$) have been selected with a model selection procedure, as described in the following (Sec. V-A).

TABLE I
Acquisition conditions. In cases 5 and 6 a one kilogram weight was tied to the right ankle to simulate an injury.

| Condition | Speed | Shoes? | Weight? |
|---|---|---|---|
| Case 1 | 2 Km/h | yes | no |
| Case 2 | 2 Km/h | no | no |
| Case 3 | free,fixed | yes | no |
| Case 4 | free,fixed | no | no |
| Case 5 | free,fixed | yes | yes |
| Case 6 | free,fixed | no | yes |
| Case 7 | variable, pre-defined | yes | no |

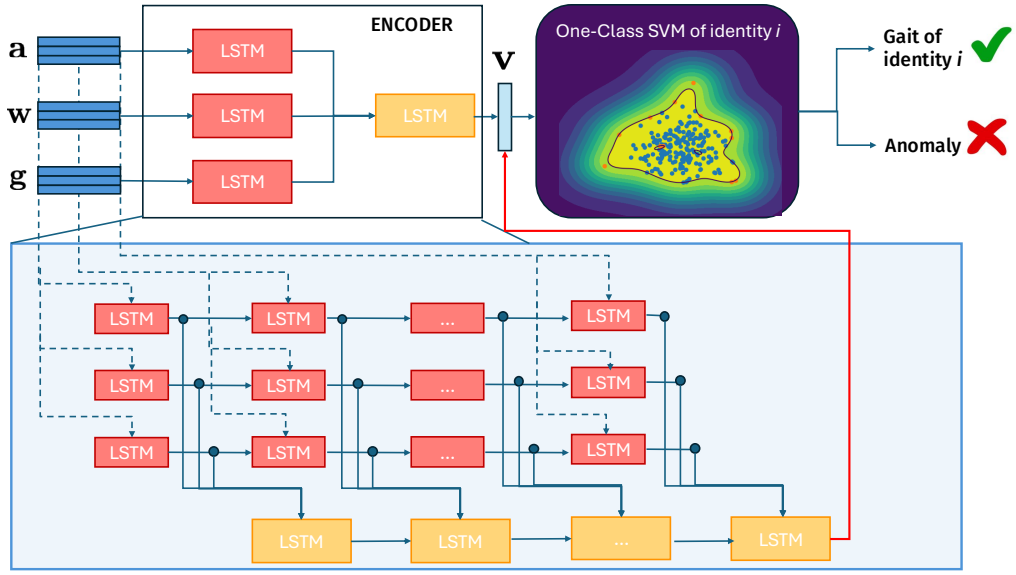
## IV. Dataset and gait segmentation

In this work we consider a dataset acquired in-house, the MARCH dataset[1], which includes inertial data of 50 participants with no motor impairment, gender-balanced (26 males and 24 females), of age $26.8 \pm 2.9$, acquired while walking on a treadmill. Inertial data have been acquired with a mobile phone, a Samsung Galaxy S25, placed in the left back pocket of participants' trousers. The acquisitions have been made in 7 controlled conditions, as summarized in Tab. I, to simulate different challenges and increase data variability. The duration of each acquisition (one for each participant in each condition) was set to 5 mins with acquisition frequency 50Hz, and all participants were asked to walk naturally on the treadmill. In cases 3–6, participants have been asked to select a comfortable speed in the range $[2, 5]$ Km/h.

The mobile phone recorded acceleration $[m/s^2]$, angular velocity $[rad/s]$, and gravity $[m/s^2]$ projections onto three axes. We segment the data into individual gait cycles, which serve as inputs for the subsequent computational steps. Although segmentation could in principle be performed directly on the data, in this initial study we relied on an auxiliary IMU

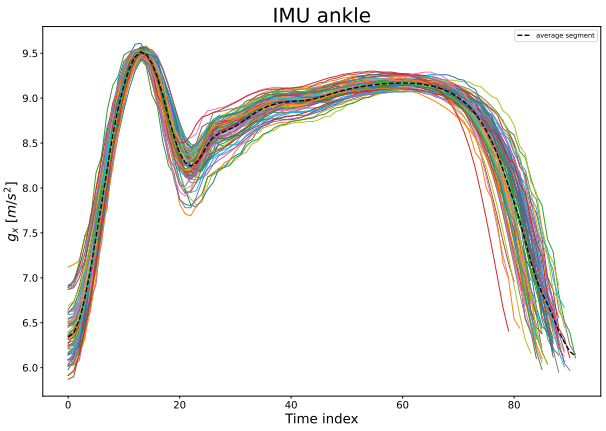
[1]The process to publish the dataset is currently ongoing; before the end of the publishing phase,it will be made available upon request

Fig. 1. A sketch of our pipeline. We address user authentication as an anomaly detection task. We learn an unsupervised gait representation using an LSTM-based encoding, and then input such representations to a One-Class Support Vector Machine (OCSVM). While the unsupervised gait representation is common, a OCSVM model is trained for each identity to be identified.




sensor placed on the participants' right ankle to improve segmentation accuracy, based on the following considerations. During the swing phase of a gait cycle, the ankle trajectory can be modelled as a cycloid, that is, the curve traced by a point on the rim of a circle rolling without slipping along a straight line. Along this trajectory, the projection of gravity onto the x-axis (the axis perpendicular to the ground) in the IMU reference frame reaches a maximum during mid-stance and mid-swing, and a minimum during pre-swing and terminal swing. Therefore, we segment each sequence into gait cycles by detecting these local minima. Fig.2 shows examples of segmentations of gravity (x component in the plot) we obtained, with the black dashed line representing the average segment. The time instants marking the segments are used to segment the sequences acquired with the mobile phone. The final dataset includes a total of 63316 gait segments, with an average of $\sim 9000$ segments for each case, and an average of $\sim 1270$ segments for each participant. Before processing, all segments are normalized to the range [0, 1].

Fig. 2. Gravity-based segmentation of the signal obtained from the right ankle IMU. The dashed black line represents the average (mean) signal calculated across all segments.




## V. Experimental Analysis

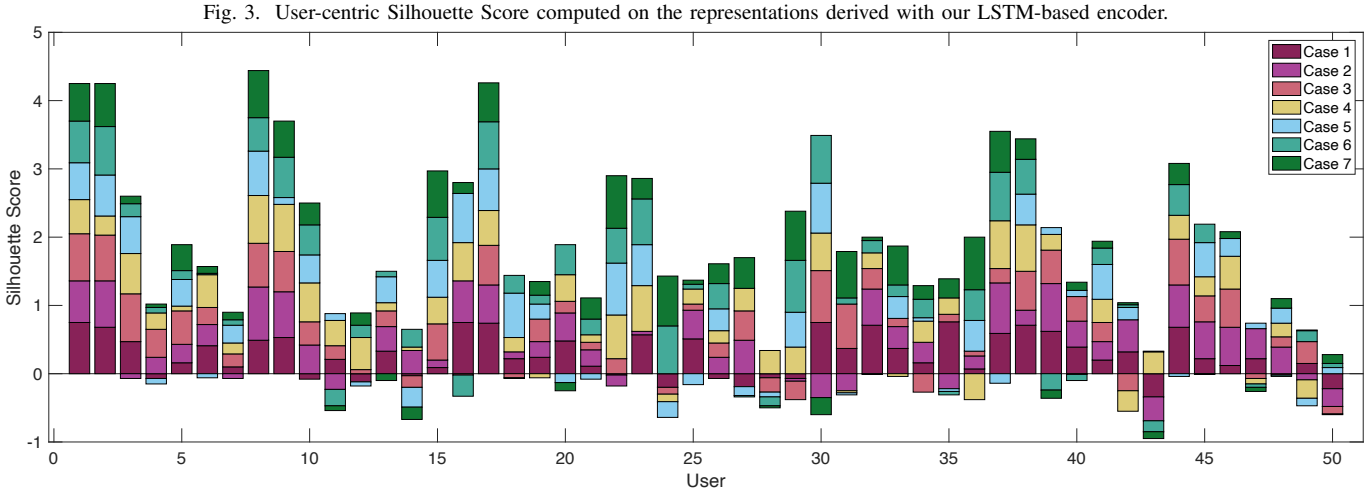
### A. Data splitting, training/test strategy and model selection

Following seminal works [29] for training the Auto-Encoder we randomly sample, without replacement, 80% of the segments from every time series, leaving the remaining 20% to serve as the validation dataset. We use Mean Squared Error (MSE) as the loss function, and Adam as optimizer. We set the learning rate to 0. initially, and then decrease it by a factor of 0.5 if the validation loss stopped improving for 10 consecutive epochs. We apply early stopping (with patience=25) to prevent overfitting. We train the OCSVM using a sequential data split of 64% segments for the training set, 16% for the validation set, and the remaining 20% for the test set. During the training phase, only segments from Cases 1, 2, 3, and 4 are utilized. To perform model selection and identify the best values of hyperparameters for OCSVM, we adopt the procedure described by Algorithm 1.

In the testing phase to evaluate the model robustness:

- From cases 1,2,3 and 4 we derive a baseline under normal conditions
- From cases 5 and 6, we study the influence of external factors, such as a variation in walking style
- From case 7, we study the influence of variable speeds and walking surface slope.

Fig. 3. User-centric Silhouette Score computed on the representations derived with our LSTM-based encoder.




**Algorithm 1** Selection of the optimal OCSVM parameter and threshold for identity $i$

**Require:** Training set $T_i$ and validation set $V_i$ of identity $i$, regularization parameters $\mathcal{N} = \{\nu_1, \ldots, \nu_{|\mathcal{N}|}\}$, and threshold range $\mathcal{P} = \{\rho_1, \ldots, \rho_{|\mathcal{P}|}\}$

**Ensure:** Optimal parameter $\nu^\star$, model $m^\star$, and threshold $\rho^\star$

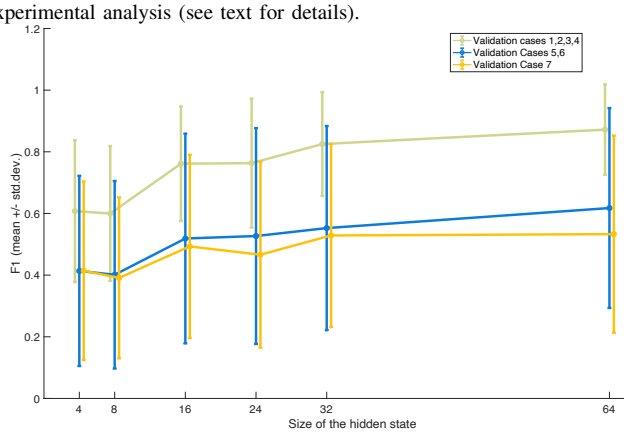
$\mathcal{M} \leftarrow \emptyset$, $\boldsymbol{\rho} \leftarrow \emptyset$, $\mathbf{F1} \leftarrow \emptyset$
**for** $j = 1, \ldots, |\mathcal{N}|$ **do**
  $m_j \leftarrow \text{trainOCSVM}(T_i, \nu_j)$
  $\rho_j, F1_j \leftarrow \text{findBestThreshold}(m_j, V_i, \mathcal{P})$
  Store $m_j$ in $\mathcal{M}$, $\rho_j$ in $\boldsymbol{\rho}$, and $F1_j$ in $\mathbf{F1}$
**end for**
$j^\star \leftarrow \arg\max_j F1_j$
$\nu^\star \leftarrow \nu_{j^\star}$
$m^\star \leftarrow \mathcal{M}_{j^\star}$
$\rho^\star \leftarrow \boldsymbol{\rho}_{j^\star}$
**return** $\nu^\star$, $m^\star$, $\rho^\star$

### *B. Latent representation analysis*

We analyse the performance of the OCSVM models on the validation sets to the purpose of selecting an appropriate size for the hidden state of the global LSTM-based encoder, which is also the latent representation we adopt for training the one-class models. We compute the F1 score for each individual model, trained according to the procedure described in Algorithm 1, and we report in Fig. 4 the average F1 obtained for the authentication tasks considering the three test scenarios described in Sec. V-A. From the curves, a dimension of 32 appears a good choice for our purposes, and it will be adopted in the remainder of the experimental analysis.

To assess the intrinsic value of the latent representation before the downstream task is applied, we may resort to an unsupervised learning metrics, the Silhouette score. This score ranges from -1 (indicating overlapping and sparse clusters) to +1 (indicating dense and well-separated clusters), taking into account both the cohesion within a cluster and the separation between different clusters. Figure 3 provides an overall impression of the scores we obtained on the partition of gait segments grouped by individual. The plot clearly highlights the heterogeneity across individuals, with some users exhibiting well-defined clusters (e.g., users 8 and 17), and others with noisier representations (e.g., user 43). Despite being low, the median scores for all cases are positive (from the first one, 0.22, 0.29, 0.23, 0.30, 0.2, 0.19, 0.17) showing an increasing complexity, as expected, on the last cases. Notice the effect of the shoes: the scores are higher in cases where users were not wearing shoes (e.g. 2 and 4).

Fig. 4. An analysis of the influence of the size of $\mathbf{v}$, our gait representation, on the downstream gait-based authentication tasks. The plot reports the performance on the validation sets of the 3 test scenarios considered in our experimental analysis (see text for details).




### *C. Results on the authentication tasks*

We now analyse more in depth the performances on the downstream authentication tasks. Figure 5 reports the F1 performance of all users on the three test scenarios, corresponding to the average scores reported in the first row of Tab. II. Overall, the authentication methodology provides a good performance in the baseline evaluation (Cases 1–4). However, the

model's response to anomalous conditions reveals interesting user-specific variations. For a subset of 15 participants, the system achieves higher F1 scores during either the impairment or dynamic speed evaluations than in the baseline scenario. Specifically, 7 users exhibits optimal performance under the simulated impairment condition (Cases 5–6), while another 7 surprisingly achieve their highest scores during the variable-speed trials (Case 7). Furthermore, 15 users show greater resilience to dynamic speed variations than to physical impairment. Finally, two distinct outliers (Users 40 and 44) have been identified, exhibiting consistently poor classification performance across all evaluation scenarios, which require further investigation into their specific gait characteristics.

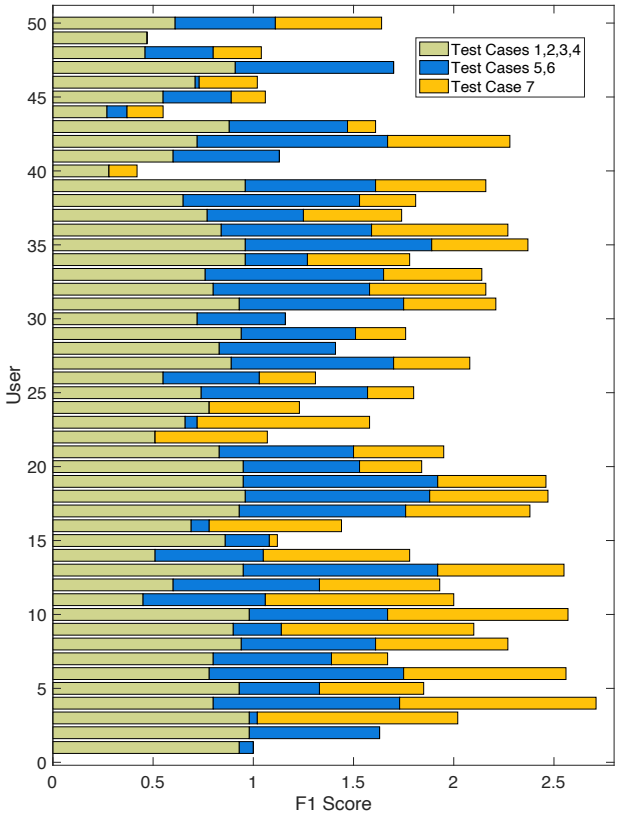
We report in Table II an overall comparison on the use of different encoders. In addition to F1, we also report the False Acceptance Rate (FAR), commonly adopted for evaluating authentication tasks. We consider two different fusion strategies – early fusion of input features, or a middle fusion (see Fig. 6) – and a different auto-encoding architecture of complexity comparable to the LSTM-based one, adopting convolutional layers. For early fusion, we stack the inertial features sequences – one feature for channel – and resample all sequences to a common length $M$, obtaining an input size $3 \times M \times 3$ (x-y-z components $\times$ sequence length $\times$ inertial features. For the LSTM-based encoder with early fusion we simply provide the stacked features in input to an LSTM layer with hidden state of size 32 which also serves as a final sequence encoding. As for the convolutional architectures, we design the following simple structures. For the early fusion, we adopt a 2D convolutional layer (with 16 channels of size $3 \times 3$) followed by max-pooling layer. The output is then flattened and embedded into a $d$-dimensional latent space using a sequence of dense layers of size 560, 35, 17, d. For middle fusion, each features sequence is first processed in parallel using a 1D convolutional layer (each with 16 channels of size $3 \times 1$). The resulting feature vectors are then stacked and fed to 2D convolutional layer (with 16 channels of size $3 \times 3$). Finally, the output is flattened and embedded into a $d$-dimensional latent space using a sequence of dense layers of size 1056, 500, 250, d.

In general, the LSTM-based architectures consistently outperform the CNN baselines, particularly during the more challenging evaluation scenarios involving simulated impairment and dynamic speed variations (Cases 5–7). Furthermore, the choice of fusion strategy significantly impacts model efficacy. For the LSTM networks, intermediate fusion generated a more robust and representative latent space than early fusion across all testing conditions. Interestingly, this trend does not hold for the CNN models, where early fusion proved comparable or even superior in challenging scenarios, whereas intermediate fusion performed best under baseline conditions.

An analysis of the False Acceptance Rate (FAR) confirms the superiority of LSTM-based encoding (improved of an order of magnitude with respect to CNN-based encodings), while also suggesting the importance of mid-fusion as the data complexity increases (in test cases 5 to 7).

Ultimately, the results demonstrate that the LSTM-based auto-encoder utilizing an intermediate fusion strategy constitutes the most effective and robust pipeline for continuous gait authentication.

Fig. 5. User-centric F1 measure for LSTM-based encoder with hidden space of size 32 and mid-level feature fusion.




## VI. Conclusions

In this study, we proposed and evaluated a biometric authentication pipeline leveraging smartphone-derived inertial gait data. To optimize representation learning, we conducted an ablation study comparing multiple architectures and data fusion strategies. Our approach, based on an LSTM auto-encoder utilizing a middle fusion strategy, outperformed conventional CNN baselines. Then, an anomaly detection procedure based on OCSVM, allowed us to isolate genuine users without requiring negative-class during training. Our results speak in favour of device-centric gait authentication, also in the pres-

Fig. 6. Input features fusion strategies: with middle fusion (left) features are first processed independently and later combined, while with early fusion (right) features are stacked in a multi-channel input tensor.

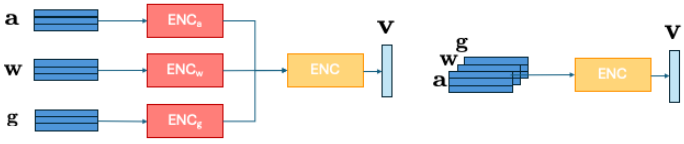


TABLE II

A COMPARISON OF OUR PIPELINE WITH ALTERNATIVE FEATURES ENCODING STRATEGIES. WE REPORT F1 (↑) AND FALSE ACCEPTANCE RATE (↓). WE MARKED IN BOLD THE BEST PERFORMANCE (MEAN AND STANDARD DEVIATION) IN EACH COLUMN.

| Encoder | Fus. | Test Cases 1–4 | | Test Cases 5–6 | | Test Case 7 | |
|---|---|---|---|---|---|---|---|
| | | F1 | FAR | F1 | FAR | F1 | FAR |
| LSTM (32) | M | $\mathbf{0.76 \pm 0.18}$ | $\mathbf{0.004 \pm 0.006}$ | $\mathbf{0.53} \pm 0.31$ | $\mathbf{0.002 \pm 0.003}$ | $\mathbf{0.44} \pm 0.28$ | $\mathbf{0.002 \pm 0.003}$ |
| LSTM (16) | M | $0.70 \pm 0.20$ | $0.005 \pm \mathbf{0.006}$ | $0.50 \pm 0.32$ | $0.004 \pm 0.008$ | $0.41 \pm 0.29$ | $0.003 \pm 0.006$ |
| LSTM (32) | E | $0.55 \pm 0.24$ | $0.02 \pm 0.04$ | $0.39 \pm 0.30$ | $0.02 \pm 0.03$ | $0.32 \pm 0.28$ | $0.01 \pm 0.03$ |
| LSTM (16) | E | $0.55 \pm 0.24$ | $0.03 \pm 0.05$ | $0.38 \pm 0.29$ | $0.02 \pm 0.04$ | $0.30 \pm 0.25$ | $0.02 \pm 0.04$ |
| Conv 1D-2D (32) | M | $0.56 \pm 0.23$ | $0.01 \pm 0.02$ | $0.33 \pm 0.29$ | $0.008 \pm 0.015$ | $0.25 \pm \mathbf{0.22}$ | $0.006 \pm 0.011$ |
| Conv 1D-2D (16) | M | $0.58 \pm 0.22$ | $0.01 \pm 0.02$ | $0.34 \pm 0.30$ | $0.009 \pm 0.021$ | $0.28 \pm 0.23$ | $0.006 \pm 0.014$ |
| Conv 2D (32) | E | $0.38 \pm 0.21$ | $0.06 \pm 0.07$ | $0.33 \pm \mathbf{0.23}$ | $0.05 \pm 0.06$ | $0.32 \pm 0.25$ | $0.04 \pm 0.04$ |
| Conv 2D (16) | E | $0.35 \pm 0.21$ | $0.09 \pm 0.09$ | $0.31 \pm 0.25$ | $0.08 \pm 0.08$ | $0.28 \pm 0.23$ | $0.05 \pm 0.05$ |

ence of variations in walking style. Ultimately, our methodology presents a highly accessible and non-intrusive solution for continuous mobile security. Nevertheless, the identification of outlier profiles with consistently poor classification metrics underscores the inherent complexities of universal biometric modelling. Future research will systematically investigate the specific kinematic anomalies of these outlier profiles to refine the feature extraction process.

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